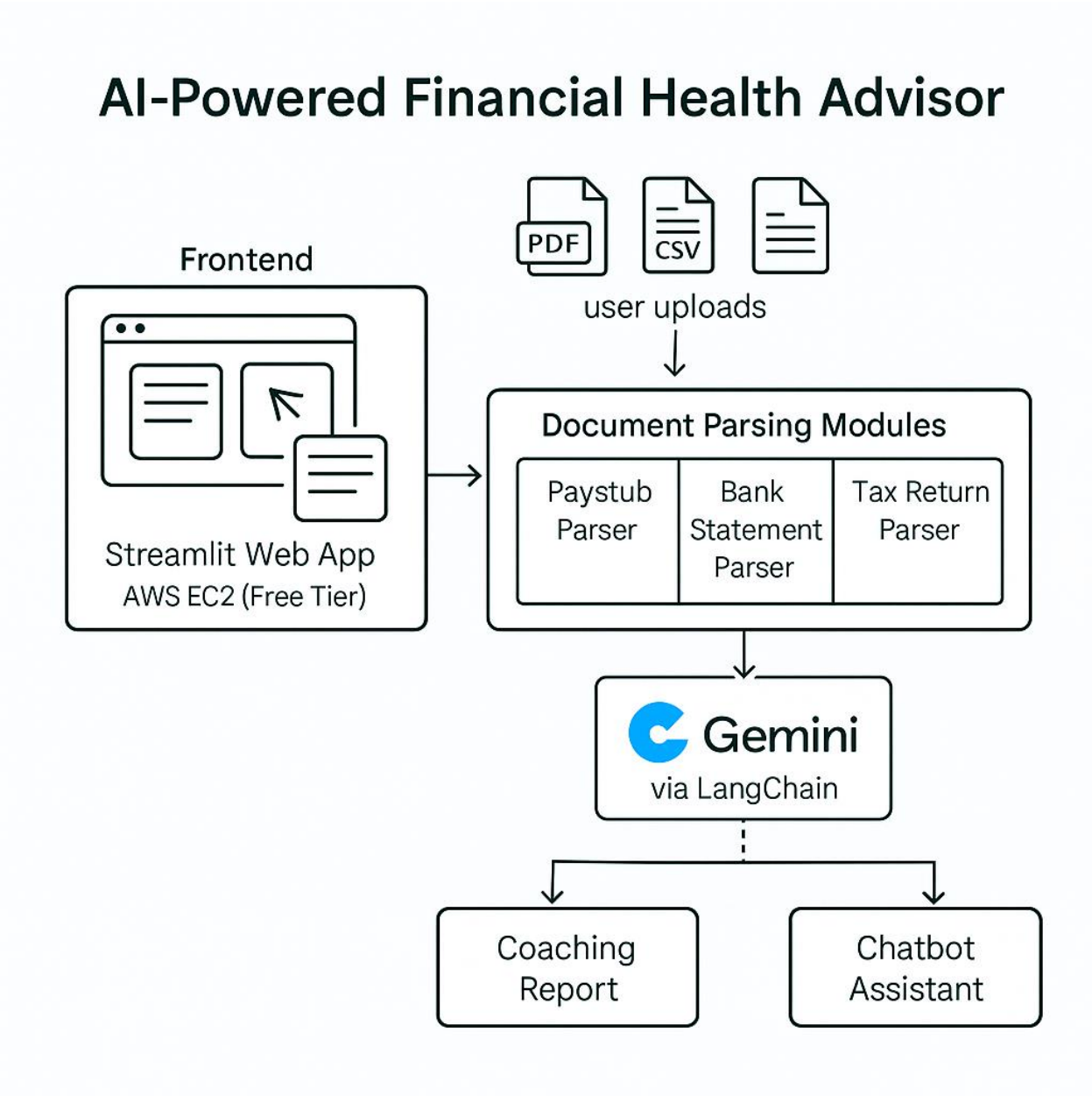
# AI Powered Financial Advisor – Capstone Project

## Project Architecture Overview

HEMACHANDRA HD

## System Architecture Description

Your system is composed of modular components that work together to process user-uploaded financial documents, extract key information, and generate personalized financial coaching using a hosted Large Language Model (Gemini 1.5 Pro).



**Frontend: Streamlit Web App**

- Built using **Streamlit**
- Hosted on **AWS EC2 (Free Tier)**
- Features:
  - Sidebar for uploading multiple documents
  - Document type detection (paystub, bank statement, tax return)
  - Tabs for:
    - Document preview
    - Extracted data
    - Coaching report
    - Chatbot interaction

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**Document Parsing Modules**

- Located in the parsers/ folder
- Three dedicated parsers:
  - `gemini_paystub_parser.py`
  - `gemini_bank_parser.py`
  - `gemini_generic_tax_parser.py`
- Uses **Gemini API via LangChain** to:
  - Extract structured fields from raw document text
  - Normalize values (e.g., earnings, deductions, tax codes)

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**LLM Integration: Gemini via LangChain**

- Hosted LLM: **Gemini 1.5 Pro** (Google)
- Accessed via **Google Generative AI SDK**
- Used in three modules:
  1. Coaching Report Generator
  2. Chatbot Assistant (via LangChain Agent)
  3. Document Field Extractor

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**LangChain Orchestration**

- Used to:
  - Handle **prompt chaining**
  - Parse **structured outputs**
  - Enable **chat memory and tool-calling**
- Ensures consistency across multi-step outputs

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**AWS Deployment**

- EC2 Ubuntu instance configured with:
  - Python + virtual environment
  - Secure API key setup
  - Port 8501 open for public access
- No LLM hosted locally — Gemini API is used via secure API call

**Tech Stack Summary**

| Layer | Tool/Service | Notes |
|---|---|---|
| Web UI | Streamlit | Lightweight, clean, and reactive UI |
| Hosting | AWS EC2 | Ubuntu instance (Free Tier) |
| LLM | Gemini 1.5 Pro | Used via Google Generative AI API |
| Orchestration | LangChain | Prompt chaining, tool calling, structured parsing |
| Parsing | PyMuPDF, Gemini | Raw text extraction + LLM field identification |
| Backend Logic | Python | Parsers, utilities, report generator |
| Docs | PDF, CSV, Plain Text | Input format support |